IFP22290120A-96Ah 锂离子电池

产品规格书

Product Specification of

IFP22290120A-96Ah Lithium-ion Rechargeable Cell

制　定 Formulated by		审　核 Checked by	批　准 Approved by
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前　言 Preface

本标准为公司统一执行的企业标准。

The specification is the enterprise standard uniformly implemented by the company.

本标准的编写格式符合 GB/T 1.1-2009《标准化工作导则 第 1 部分：标准的结构和编写》的规定。

The format of this specification is in accordance with the provisions of GB/T 1.1-2009, standardization work guideline Part 1: Standards Structure and Compilation.

本标准在参照：GB/T 31484-2015《电动汽车用动力蓄电池循环寿命要求及试验方法》、GB 38031-2020 《电动汽车用动力蓄电池安全要求》、GB/T 31486-2015 《电动汽车用动力蓄电池电性能要求及试验方法》、IEC62619-2017、IEC 62660-2-2018、GB/T36276-2018、Q/GX 60013-2020《电动汽车用锂离子动力电池单体电性能测试方法》的基础上，结合我公司产品实际和试验条件，特制定《IFP3130093-99Ah 锂离子电池产品规格书》标准，并对试验方法、判定标准内容进行了阐述，以指导 IFP3130093-99Ah 锂离子电池产品的性能检测。

This specification refers to GB/T 31484-2015 cycle life requirements and test methods for power cell for electric vehicles, GB 38031-2020 safety requirements and test methods for power cell for electrical vehicles, GB/T 31486-2015 electrical performance requirements and test methods for power cell for electric vehicles, IEC62619-2017, IEC 62660-2-2018, GB/T36276-2018, Q/GX 60013-2020 electrical performance test method of lithium ion power cell for electrical vehicles. Combined with the actual and test conditions of our company's product, the standard of product specification for IFP3130093-99Ah lithium ion rechargeable cell is specially formulated. The test method and criteria is revised and supplemented to guide the manufacturing and approval of IFP3130093-99Ah lithium ion cell.

注：若修改，以最新发布标准为准。

Remark：If modified, please take the latest version as standard.

修订记录 Modification Record

版本号 Version Number	变更内容 Description of Changes	修订日期 Revision date	修订者 Reviser
	首次发布 1st Edition	2024.01.20	高程 GAO Cheng

术语和定义 Terms and Definition

产　　品：本规格书中的 "产品" 是指合肥国轩高科动力能源有限公司 (以下简称国轩) 生产的 IFP22290120A-96Ah 可充电方形铝壳锂离子电池。

Product: Refers to the IFP22290120A-96Ah rechargeable lithium-ion cell with prismatic aluminum shell manufactured by Gotion high-tech power energy Co., Ltd. (hereinafter referred to as Gotion) in this specification.

客　　户：指与国轩签署产品销售合同中的买方。

Customer: Refers to the buyer in the product sales contract signed with Gotion.

环境温度：电池所处的周围环境温度。

Environment temperature: The ambient temperature where the cell is located.

电池温度：由接入电池表面中心的温度传感器测量的电池表面的温度，温度传感器和测量线路的选择由国轩和客户共同商定。

Cell temperature: The temperature measured by temperature sensor installed at the center of cell surface. The selection of temperature sensor and measuring line shall be jointly agreed by Gotion and the customer.

新鲜电池：指电池自产品生产完成日期算起 7 天以内的状态。

Fresh cell: Refers to cell within 7 days after production.

充电倍率：充电电流与电池管理系统监测的电池的容量值的比率。例如：电池容量为 96 Ah，充电电流为 48 A 时，则充电倍率为 0.5C；当电池容量跌落为 90 Ah，充电电流为 45 A 时，则充电倍率为 0.5C。

Charging Rate: The ratio of the charging current to the capacity which measured by the battery management system. For example, if the cell capacity is 96 Ah and the charging current is 48 A, the charging rate is 0.5C. If the cell capacity drops to 90 Ah and the charging current is 45 A, the charging rate is 0.5C.

荷电状态：在无负载的情况下，以安培小时或者以瓦特小时为单位计量的电池容量状态与标称容量的比值，缩写用 SOC 表示。 例如：若将容量为 96 Ah 的状态视为 100% SOC，若容量为 0 Ah 时， SOC 为 0%。

State of charge: Under unloaded conditions, the ratio of the cell capacity state to the nominal capacity measured in ampere-hour or watt- hour. The abbreviation is expressed by SOC. For example, if the capacity at 96 Ah considered as 100% SOC, the capacity at 0 Ah, considered as 0% SOC.

健康状态：电池实际容量与标称容量的比值，缩写用 SOH 表示。 例如： 电池容量 96Ah 为 100% SOH，容量衰减为 76.8Ah 时，SOH 为 80%。

State of health: The ratio of actual cell capacity to nominal capacity, the abbreviation is expressed by SOH. For example, if the cell capacity at 96 Ah considered as 100% SOH, the cell capacity decays to 76.8 Ah, considered as

80% SOH.

循　　环：电池按规定的充放标准充放一次为一个循环。循环包括短时的正常充电或者再生充电和放电过程的组合，在充电过程中有时只有正常充电而无再生充电的情况，放电可以由一些部分放电组合在一起形成。

Cycle: The cell shall be charged and discharged once according to the specified charging and discharging standards as a cycle. The cycle includes short-term normal charging or a combination of regenerative charging and discharging processes. In the charging process, sometimes there is only normal charging and no regenerative charging. The discharge can be formed by combining some partial discharges.

标准充电：本规格书第 4.1.1 条所述的充电模式。

Standard charge: The charging mode described in 4.1.1 of this specification.

标准放电：本规格书第 4.1.2 条所述的放电模式。

Standard discharge: The discharging mode described in 4.1.2 of this specification.

开路电压：没有接入任何负载和电路时测得的电池的电压，缩写用 OCV 表示。

Open circuit voltage: The voltage of the cell measured when unloaded or circuit is disconnected. The abbreviation is expressed by OCV.

交流内阻：给电池正负极注入 1kHz 的正弦波电流，测试所得到的内阻值，缩写用 ACR 表示。

AC resistance: Apply 1kHz sine wave current between the positive and negative poles of the cell, and the internal resistance obtained, which abbreviated as ACR.

直流内阻：工作条件下电池的电压变化与相应的电流变化之比，缩写用 DCR 表示。

DC resistance: The ratio of the voltage changes to the corresponding current change under working conditions, and the abbreviation is DCR.

模　　组：锂离子电池经串并联方式组合，加装单体电池监控与管理装置后形成的电池与 pack 的中间产品。

Module: The intermediate product between single cell and pack, which is formed by lithium-ion cells in series and parallel after installing cell monitors and management devices.

脉冲电流：以周期重复出现的电流为脉冲电流，脉冲电流或是以同一方向出现，或是以正、负交替变换方向出现。

Pulse current: The currents that appear periodically are called pulse currents. The pulse currents appear either in the same direction or in alternating positive and negative directions.

压 缩 力：模组组装时，电池可承受压缩力的安全边界。

Compression force: When the module is assembled, the cell can withstand the force perpendicular to the cell stacking direction.

膨 胀 力：在使用过程中，因极片厚度反弹等固有特性导致电池膨胀，而对约束夹具产生的作用力。

Swelling force: The inherent characteristics such as the rebound of the electrode thickness may lead to cell expansion during use which generates force on the clamp.

测量单位：见下表

Units of measurement: Refer to following table

序号 No.	单位 Units	简写 Abbreviation	单位类型 Type of units
1	伏特 Volt	V	电压单位 Voltage
2	安培 Ampere	A	电流单位 Current
3	安培-小时 Ampere-Hour	Ah	容量单位 Capacity
4	瓦特-小时 Watt-Hour	Wh	能量单位 Energy
5	欧姆 Ohm	Ω	电阻单位 Resistance
6	毫欧姆 Milliohm	m Ω	电阻单位 Resistance
7	摄氏度 Degree Celsius	℃	温度单位 Temperature
8	毫米 Millimeter	mm	长度单位 Length
9	秒 Second	s	时间单位 Time
10	赫兹 Hertz	Hz	频率单位 Frequency
11	牛顿 Newton	N	力单位 Force
12	千克力 Kilogram-Force	kgf	力单位 Force

1. 基本信息 Basic Information

1.1 适用范围 Scope

本产品规格书由合肥国轩高科动力能源有限公司提供给客户，描述了其锂离子二次电池的性能。

This specification specification is provided to customers by HEFEI GUOXUAN HIGH-TECH POWER ENERGY Co. Ltd., describing Lithium ion rechargeable cell properties.

1.2 用途 Application：EV

1.3 产品类型 Product type：锂离子二次电池，方形，LFP 电芯 Lithium ion rechargeable battery

1.4 型号名称 Model Name：IFP22290120A-96Ah

2. 规格 Specification

2.1 标准技术参数 Standard Technical Parameters

项目 Item	参数 Specification	条件/注释 Condition/Notes
2.1.1 标称容量 Nominal Capacity	96Ah	参考 4.1.1 与 4.1.2 标准充放电模式测试 Refer to 4.1.1&4.1.2 standard charge and discharge procedure
2.1.2 标称电压 Nominal Voltage	3.21V	0.33C，25±2℃
2.1.3 充放电电压范围 Charge/Discharge Voltage Range	2.5 V～3.65V	T>0℃
	2.0 V～3.65V	T≤0℃
2.1.4 重量 Weight	1730±30g	
2.1.5 质量能量密度 Weight energy density	180Wh/kg	0.33C，25±2℃
2.1.6 体积能量密度 Volume energy density	400Wh/L	0.33C，25±2℃
2.1.7 交流内阻 ACR	0.3mΩ≤ACR≤0.6mΩ	25℃±2℃ 新电池状态（17%±3%SOC） Fresh cell （17%±3%SOC）
2.1.8 直流内阻 DCR	0.3mΩ≤DCR≤0.8mΩ	50%SOC，25±2℃（5C10S））

2.2 推荐充电规范 Recommended Charge specification

项目 Item	参数 Specification	条件/注释 Condition/Notes
2.2.1 常规充电（慢充）Regular Charge (Slow charge)	48A 3.65V 4.8A 10℃≤T＜45℃	恒流 Constant current 恒压 Constant voltage 截止条件 Cut off condition 温度 Temperature
2.2.2 阶梯充电（快充） Step Charge	见附件 A.1 See appendix A.1	不同温度的快充策略 Fast charging strategy at different temperature

2.3 工作温度范围 Working Temperature Range

项目 Item	参数 Specification	条件/注释 Condition/Notes
2.3.1 最佳工作温度 Optimum working Temperature	10～35℃	
2.3.2 充电温度范围 Charge temperature range	-20～55℃	
2.3.3 放电温度范围 Discharge temperature range	-35～60℃	电芯温度≤60℃ Cell temperature≤60℃

3. 外观尺寸 Appearance and Dimension

3.1 外观 Appearance

电池外表面无明显的划痕、裂痕、生锈、变色或电解液泄漏，无影响电池正常使用的其它外观缺陷。

Without obvious scratches, cracks, rust, discoloration or electrolyte leakage, no other appearance defects affecting cell normal operation

3.2 尺寸 Dimension

厚度：（22.2±0.5）mm (含外包膜厚度，17%±3%SOC，200kgf)

Thickness：(22.4±0.5) mm（including outer film thickness, 17%±3%SOC, 200kgf）

宽度： (284.8±0.5) mm（含外包膜）

Width：(284.8±0.5)mm（including outer film thickness）

宽度： (290±0.5) mm（含极柱，含外包膜和顶盖贴片）

Width：(290±0.5)mm（including outer film thickness and terminals）

高度： (120±0.5) mm（含外包膜厚度）

Height：(120±0.5)mm（including outer film thickness）

4. 性能测试规范 Performance Specification

4.1 标准测试条件 Standard Testing Condition

4.1.1 单体电池充电 Single cell charging

室温下（25℃±2℃）， 单体电池以 96A 电流放电至电压为 2.5V，静置 30min，然后再以 96A 电流充电至电压为 3.65V 时转恒压充电，至充电电流降至 4.8A 时停止充电，充电后静置 30min。

At room temperature（25℃±2℃）, 96A constant current discharge to 2.5V, rest 30min, then 96A constant current charge to 3.65V followed by constant voltage charging until current drops down to 4.8A, rest 30min.

4.1.2 单体电池放电 Single cell discharging

室温下（25℃±2℃）， 单体电池以 96A 电流放电至电压为 2.5V，静置 30min。

At room temperature（25℃±2℃）, 96A constant current discharge to 2.5V, rest 30min.

4.2　电气特性 Electrical Properties

项目 Item	参数 Specification	条件/注释 Condition/Notes
4.2.1 初始容量 Initial Capacity	≥99Ah	按照 4.1.1 充电和 4.1.2 放电 Refer to 4.1.1&4.1.2 standard charge and discharge procedure
4.2.2 容量与温度的相关性 Capacity and temperature correlation	55℃，　≥100% 初始容量 25℃，　100% 初始容量 0℃，　≥90%初始容量 -20℃，　≥80%初始容量 -30℃，　≥75% 初始容量 55℃, ≥95%Initial Capacity 25℃, 100% Initial Capacity 0℃, ≥90% Initial Capacity -20℃, ≥80% Initial Capacity -30℃, ≥75% Initial Capacity	25℃±2℃温度下按照 4.1.1 充电，在测试温度下静置 20h，以 1C 放电至 2.5V（0℃及以下 2.0V）。 Charge refer to 4.1.1, rest 20h at testing temperature, 1C discharge to 2.5V at testing temperature（ to 2.0V at or below 0℃）.
4.2.3SOC-OCV 表 SOC-OCV Table	见附件 A.4 See appendix A.4	按照 4.1.1 充电，然后以 1C 放电 5% 容量（此容量为按照 4.1.2 放电至对应截止电压（[-30℃~0℃）2.0V, [0℃~55℃] 2.5V）对应的容量，静置 X h(25℃ 2h，>25℃ 3h，<25℃ 5h)，重复 20 次，记录静置后的电压数据。Charge as per 4.1.1, then discharge for 5% capacity at 1C（The capacity is obtained by discharging to certain cut-off voltage as per 4.1.2（[-30℃-0℃）2.0V,[0℃-55℃] 2.5V），静置 X h(25℃ 2h，>25℃ 3h，<25℃ 5h)，, repeat 20 times, record voltage after rest.
4.2.4 不同温度和不同 SOC 放电电阻（DCR） Discharge DCR at different temperature and SOC	见附件 A.5 See appendix A.5	按照 4.1.1 充电，然后以 1C 按照 4.2.2 不同温度下放电容量调整 SOC（截止电压参照 4.2.3），静置 1h，测试 3C 放电 10s 的 DCR。 放电电阻为开路电压与放电末端电压的差值除以电流，测试 SOC 为 95%，90%，80%，70%，60%，50%，40%，30%，20%，10%。 Charge as per 4.1.1, then 1C Discharge to adjust SOC according to the 4.2.2 discharge capacity at different temperatures (refer to 4.2.3 for cut-off voltage), rest 1h, and test the DCR of 3C discharge for 10s. The discharge

		resistance is the difference between the open circuit voltage and the discharge terminal voltage divided by the current. The test SOC is 95%, 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 10% .
4.2.5 不同温度和不同 SOC 充电电阻（DCR） charge DCR at different temperature and SOC	见附件 A.6 See appendix A.6	按照 4.1.1 充电，然后以 1C 按照 4.2.2 不同温度下放电容量调整 SOC（截止电压参照 4.2.3），静置 1h，测试 2.25C 充电 10s 的 DCR。充电电阻为充电末端电压与开路电压的差值除以电流，测试 SOC 为 5%，10%，20%，30%，40%，50%，60%，70%，80%，90%。 Charge as per 4. 1. 1, then 1C Discharge to adjust SOC according to the 4.2.2 discharge capacity at different temperatures (refer to 4.2.3 for cut-off voltage), rest 1h, and test DCR charged at 2.25c for 10s. The charging resistance is the difference between the charging terminal voltage and the open circuit voltage divided by the current. The test SOC is 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%.

4.3 最大脉冲功率 （极限能力值）Maximum plus power(Ultimate capacity value）

项目 Item	参数 Specification	条件/注释 Condition/Notes
4.3. 1 不同温度和不同 SOC 脉冲放电峰值功率 Peak power plus discharge at different temperature and SOC	见附件 A.7 See appendix A.7	按 4.2.4 测试数据计算的 DCR 和最大脉冲放电电流，计算出在不同温度规定的放电下限电压时的脉冲放电功率。 According to the DCR and the maximum pulse charging current calculated from the test data of 4.2.4, the pulse discharge power at the lowest limit voltage specified at different temperature.
4.3.2 不同温度和不同 SOC 脉冲回馈峰值功率 Peak power plus feedback at different temperature and SOC	见附件 A.8 See appendix A.8	按 4.2.5 测试数据计算的 DCR 和最大脉冲充电电流，计算出在充电上限电压 3.65V 时的脉冲回馈功率。 According to the DCR and the maximum pulse charging current calculated from the test data of 4.2.5, the pulse feedback power at the upper charging voltage of 3.65v was calculated.

4.4 耐久性能 Durability Performance

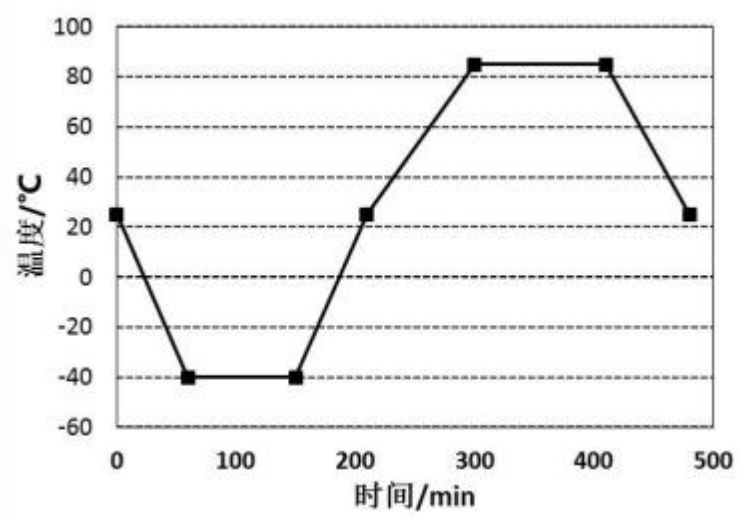
项目 Item	参数 Specification	条件/注释 Condition/Notes
4.4.1 室温荷电保持率 Room Temperature SOC Retention Rate	≥97%	25℃， 100%SOC，28days
4.4.2 室温容量恢复率 Room Temperature Capacity Recovery Rate	≥98%	
4.4.3 高温荷电保持率 High Temperature SOC Retention Rate	≥96%	45℃， 100%SOC，28days
4.4.4 高温容量恢复率 High Temperature Capacity Recovery Rate	≥97%	
4.4.5 高温荷电保持率 High Temperature SOC Retention Rate	≥95%	55℃， 100%SOC，7days
4.4.6 高温容量恢复率 High Temperature Capacity Recovery Rate	≥96%	
4.4.7 常温循环寿命 Room Temperature Cycle Life	3000 cycles	使用企业钢板夹具约束单体蓄电池，初始夹紧力 150±50kgf 25±2℃, 96A 阶梯充/96A 放，100DOD,80%EOL； 25±2℃, cycle test by the 96A step charge and 96A discharge under 150±50kgf preload
4.4.8 高温循环寿命 High Temperature Cycle Life	1500 cycles	使用企业钢板夹具约束单体蓄电池，初始夹紧力 150±50kgf 45℃, 96A 阶梯充/96A 放，100DOD,80%EOL； 45℃， cycle test by the 99A step charge and 99A discharge under 150 ± 50kgf preload
4.4.9 常温快充循环寿命 Room Temperature Cycle Life	2000 cycles	使用企业钢板夹具约束单体蓄电池，初始夹紧力 150±50kgf 25±2℃, 288A（3C）阶梯充/96A 放，100DOD,80%EOL； 25±2℃, cycle test by the 288A（3C） step charge and 96A discharge under 150±50kgf preload
4.4.10 日历寿命 Calendar Life	10 years	25±2℃， 50%SOC ,80%EOL

4.5 安全测试规范 Specification of Safety Test

项目 Item	参数 Specification	条件/注释 Condition/Notes
4.5.1 过放电 Over discharging	不爆炸，不起火，不漏液 No explosion, no fire, no leakage	单体蓄电池（使用企业钢板夹具约束单体蓄电池，夹紧力 150±50kgf）按 4.1.1 方法充电，单体蓄电池以 1 I_1 (A)电流放电 90min，并观察 1h。（I_1=99A）; The single cell was charged according to the method of 4.1.1, and the single cell was discharged with a current of 1 I1 (A) for 90 min and observed for 1 h.
4.5.2 过充电 Over charging	不爆炸，不起火 No explosion, no fire	单体蓄电池（使用企业钢板夹具约束单体蓄电池，夹紧力 150±50kgf）按 4.1.1 方法充电，以 1 I_1 (A)电流恒流充电至电压达到企业技术条件中规定的充电终止电压的 1.1 倍或 115%SOC 后停止充电，并观察 1h。（I_1=99A） The single cell is charged according to the method of 4.1.1, and is charged with a constant current of 1 I1 (A) until the voltage reaches 1.1 times of the charging end voltage specified in the technical conditions of the enterprise or the charging time reaches 1 h, and the charging is stopped, and 1 h is observed.
4.5.3 短路 Short circuiting	不爆炸，不起火 No explosion, no fire	单体蓄电池（使用企业钢板夹具约束单体蓄电池，夹紧力 150±50kgf）按 4.1.1 方法充电，将单体蓄电池正、负极经外部短路 10min，外部线路电阻应小于 5mΩ，并观察 1h。The single cell is charged according to the method of 4.1.1, and the cathode and anode terminals of the single cell are externally short-circuited for 10 min, and the external line resistance should be less than 5 mΩ, and observed for 1 h.
4.5.4 加热 Heating	不爆炸，不起火 No explosion, no fire	单体蓄电池按 4.1.1 方法充电，温度箱按照 5℃/min 的速度由室温升至 130℃±2℃，并保持此温度 30min 后停止加热，并观察 1h。The single battery was charged according to the method of 4.1.1, and the temperature box was raised from room temperature to 130 °C ± 2 °C at a rate of 5 ° C / min. After maintaining this temperature for 30 min, the heating was stopped and observed for 1 h.
4.5.5 挤压 Crushing	不爆炸，不起火 No explosion, no fire	单体蓄电池按 4.1.1 方法充电； a) 挤压方向：垂直于蓄电池极板方向施压； b) 挤压板形式：半径为 75mm 的半圆柱体，半圆柱体的长度(L)大于被挤压电池的尺寸；挤压速度：（2±1）mm/s； c) 挤压程度：电压达到 0V 或变形量达到 15%或挤压力达到 100KN 后停止挤压，保持 10min；并观察 1h。 he single cell is charged according to the method of 4.1.1; a) extrusion direction: apply pressure perpendicular to the battery plate; b) extruded plate form: a semi-cylindrical body with a radius of 75 mm, the length (L) of the semi-cylindrical body is larger than the size of the battery to be squeezed;extrusion speed: (2 ± 1) mm / s; c) degree of extrusion: after the voltage reaches 0V or the deformation amount reaches 15% or the pressing force reaches 100KN, the extrusion is stopped and kept for 10 minutes;And observe 1h.

4.5.6 温度循环 Temperature Cycle Test	不爆炸，不起火，不漏液 No explosion, no fire, no leakage	单体蓄电池按 4.1.1 方法充电，单体蓄电池放入温度箱中，温度箱温度按照下表进行调节，循环次数 5 次，观察 1h。 The single cell is charged according to the method of 4.1.1. The single cell is placed in the temperature box, and the temperature of the temperature box is adjusted according to the following table. The number of cycles is 5 times and observed for 1 hour.

温度 Temp ℃	时间增量 Time increasing min	累计时间 cumulative time min	温度变化率 Temperature change rate℃/min
25	0	0	0
-40	60	60	13/12
-40	90	150	0
25	60	210	13/12
85	90	300	2/3
85	110	410	0
25	70	480	6/7



5.使用注意事项 Notes

锂离子充电电池的使用警告。操作不当可能会造成电池发热、起火和性能劣化，务必认真阅读以下条款。
Warning on the use of rechargeable lithium-ion batteries. Improper operation may cause the battery to heat up, catch fire and deteriorate its performance. Be sure to read the following terms carefully.

5.1 注意事项 Precautions

- 应用配有电池的设备时，使用前请参阅用户手册。
  When using a device equipped with a battery, refer to the user manual before use.
- 包装前请检查正极和负极端子的方向。
  Check the orientation of the cathode and anode terminals before packaging.
- 端子或导线与电池模块相连，注意绝缘防止短路。
  The terminal or wire is connected to the cell module, pay attention to the insulation to prevent short circuit.
- 长期不用时，电池要存放于阴凉干燥处 (≤35°C，30%~50%SOC，环境湿度≤85%ROH，无凝露，3 个月进行一次充放电) 。
  Store the cell in a cool dry place (≤35°C, 30%~50%SOC，Storage ambient humidity < 85% ROH, no Condensation, charge and discharge once every 3 months) when not in use for a long time.
- 不要将电池放置于阳光直射处或热源。
  Do not place the cell in direct sunlight or heat source to prevent high temperature of the cell.
- 处理电池单元时请勿穿戴金属饰品 (如戒指，手表，饰件等) 。
  Do not wear metal accessories (such as rings, watches, accessories, etc.) when handling cell units.,
- 请勿将电池放置在本文件规定的使用温度范围之外的地方。
  Do not place the cell outside the operating temperature range specified in this document.

5.2 禁止事项 Prohibited Matters

- 请勿超过最大充电倍率充电。
  Do not charge more than the maximum charge rate.
- 请勿拆卸或改装电池。
  Do not disassemble or modify the Cell.
- 请勿抛掷或撞击电池。
  Do not throw or hit the battery.
- 请勿用锐器刺穿电池。(例如钉子，刀子，笔， 电钻)
  Do not pierce the battery with a sharp object. (eg nails, knives, pens, electric drills)
- 请勿与其它型号电池或模块单元混用。
  Do not mix with other cell or module units.
- 勿将新、旧电池在 PACK 中同时使用。
  Do not use both new and old cell in PACK.
- 请勿将电池放置于大于 60℃高温处。
  Do not place the cell at a temperature higher than 60 ℃.

- 不要将电池放入微波炉或高压容器中。

  Do not put the cell in a microwave or high pressure container.

- 请勿用导电材料连接正、负极端子。(例如金属，电线)

  Do not connect the positive and negative terminals with conductive materials. (eg metal, wire)

- 不允许弄湿或将电池浸入水或海水中。

  Do not allow to wet or immerse the cell in water or sea water.

- 请勿以制造商书面协议之外的方式使用电池。

  Do not use the battery in a manner other than the manufacturer's written agreement.

5.3 产品寿命终止管理 Product End-life Management

电池使用期限是有限的，客户应建立有效的跟踪系统监测并记录每个使用期限内电池的内阻和容量。内阻及容量的测量方法和计算方法需要客户与国轩共同讨论和双方同意。当使用中电池的内阻超过这个电池最初内阻的 150%或容量小于标称容量的 70%，应停止使用电池。违反该项要求，将免除国轩依据产品销售协议以及本规格书所应承担的产品质量保证责任及由此引起的损失赔偿等一切相关责任。

The cell life is limited. Customers should establish an effective tracking system to monitor and record the internal resistance and capacity of each cell during its life. The measurement method and calculation method of internal resistance and capacity need to be discussed and agreed between the customer and Gotion. When the internal resistance of the cell in use exceeds 150% of the initial internal resistance of the cell, or the capacity is less than 70% of the nominal capacity, the cell should not to be operated. Violation of this requirement will exempt Gotion from its responsibility for product quality assurance in accordance with the product sales agreement and this specification, as well as all related liabilities such as loss compensation caused thereby.

5.4 保密协议 Confidentiality Agreement

客户应对合作内容高度保密，未经国轩许可，不得向第三方透露规格书的任何内容，否则，将依照相关法律追究责任。

The customer shall keep the cooperation content highly confidential. Without the permission of Gotion, the customer shall not disclose any content of the technical agreement to a third party. Otherwise, the customer will be held responsible according to relevant laws.

5.5 免责声明

如果由于产品需求单位或使用者不按本说明书中的规定进行使用，国轩不再承担产品质量保证责任及由此引起的损失赔偿等一切相关责任。因前述行为对国轩的声誉造成负面影响的，国轩保留追究产品需求单位法律责任的权利。

If the product demand party or user does not use the product in accordance with the provisions of this specification, Gotion will no longer take any relevant responsibility such as product quality assurance liability and loss compensation caused thereby. In case of any negative impact on Gotion's reputation due to the above-mentioned actions, Gotion reserves the right to investigate the legal liability of the product demand part

6. 外形图纸 Mechanical Drawing



电芯整体尺寸 Cell Dimension

注：图中尺寸单位为毫米（mm）

Remark：The dimensions are shown in millimeters (mm)

附录 Appendix

A.1 阶梯充电矩阵表 Step charge Table

	-20℃ < T ≤ -10℃	-10℃ < T ≤-5℃	-5℃ < T ≤0℃	0℃ < T≤ 10℃	10℃ < T≤ 15℃	15℃ < T≤ 20℃	20℃ < T < 25℃	25℃≤T≤ 40℃	40℃ < T≤ 45℃	45℃ < T≤ 50℃	50℃ < T≤ 55℃
0%	0.150	0.200	0.300	0.500	1.000	1.500	2.000	3.000	1.500	1.000	0.500
0% ＜SOC≤5%	0.150	0.200	0.300	0.500	1.000	1.500	2.000	3.000	1.500	1.000	0.500
5% ＜SOC≤10%	0.150	0.200	0.300	0.500	1.000	1.500	2.000	3.000	1.500	1.000	0.500
10% ＜SOC≤15%	0.150	0.200	0.300	0.500	1.000	1.500	2.000	3.000	1.500	1.000	0.500
15% ＜SOC≤20%	0.150	0.200	0.300	0.500	1.000	1.500	2.000	3.000	1.500	1.000	0.500
20% ＜SOC≤25%	0.100	0.150	0.300	0.500	1.000	1.500	1.800	3.000	1.300	0.800	0.500
25% ＜SOC≤30%	0.100	0.150	0.300	0.500	1.000	1.500	1.800	3.000	1.300	0.800	0.500
30% ＜SOC≤35%	0.080	0.100	0.200	0.500	0.800	1.200	1.800	2.800	1.200	0.800	0.500
35% ＜SOC≤40%	0.080	0.100	0.200	0.500	0.800	1.200	1.800	2.800	1.200	0.800	0.500
40% ＜SOC≤45%	0.080	0.100	0.150	0.400	0.800	1.200	1.500	2.600	1.000	0.800	0.500
45% ＜SOC≤50%	0.080	0.100	0.150	0.400	0.800	1.200	1.50	2.600	1.000	0.800	0.500
50% ＜SOC≤55%	0.050	0.050	0.150	0.400	0.800	1.200	1.300	2.200	1.000	0.500	0.500
55% ＜SOC≤60%	0.050	0.050	0.150	0.400	0.800	1.200	1.300	2.200	1.000	0.500	0.500
60% ＜SOC≤65%	0.040	0.050	0.100	0.200	0.800	1.000	1.300	2.000	0.800	0.500	0.500
65% ＜SOC≤70%	0.040	0.050	0.100	0.200	0.800	1.000	1.300	2.000	0.800	0.500	0.500
70% ＜SOC≤75%	0.040	0.050	0.100	0.200	0.800	1.000	1.000	1.800	0.800	0.500	0.500
75% ＜SOC≤80%	0.040	0.050	0.100	0.200	0.800	1.000	1.000	1.800	0.800	0.500	0.500
80% ＜SOC≤85%	0.040	0.040	0.050	0.100	0.800	0.800	0.800	1.000	0.800	0.500	0.300
85% ＜SOC≤90%	0.040	0.040	0.050	0.100	0.500	0.800	0.800	0.800	0.800	0.300	0.300
90% ＜SOC≤95%	0.040	0.040	0.050	0.100	0.250	0.500	0.500	0.500	0.500	0.300	0.300
95% ＜SOC≤97%	0.040	0.040	0.050	0.100	0.250	0.500	0.500	0.500	0.500	0.300	0.300
97% ＜SOC≤98%	0.040	0.040	0.050	0.100	0.100	0.250	0.250	0.250	0.250	0.100	0.100
98% ＜SOC≤100%	0.040	0.040	0.050	0.100	0.100	0.250	0.250	0.250	0.250	0.100	0.100

A.2 单体电芯故障阈值 Single Cell failure Threshold (Recommended Value)

A.2.1 单体电芯故障阈值 Single Cell Failure Threshold (BMS)

电池型号 Cell Model	数值 Numeric value	解释 Description	参考 Reference
单体过电压严重报警阈值 Single cell overvoltage critical alarm threshold V	3.8	充电时过电压超过报警阈值，将影响电池的循环寿命，用户充电不允许超过此电压值 When the overvoltage exceeds the alarm threshold during charging, it will affect the cycle life of the cell. User charging is not allowed to exceed this voltage value.	必须强制断电 Power must be turned off
单体过电压一般报警阈值 Single cell overvoltage critical alarm threshold V	3.7	充电时过电压的临界值 Critical value of overvoltage during charging	超过 5s 强制断电 Forced power off for more than 5S
最大工作电压 Maximum working voltage V	3.65	正常工作的最大极限值 Maximum limit of normal operation	
单体欠压严重报警阈值 Maximum working voltage V	[-30℃~0℃） 1.8 [0℃~55℃] 2.0	放电时欠压超过报警阈值，将影响电池的循环寿命，用户放电不允许超过此电压值 When the under-voltage exceeds the alarm threshold during discharge, it will affect the cycle life of the cell. The user discharge must not exceed this voltage value	
单体欠压一般报警阈值 Single under voltage critical alarm threshold V	[-30℃~0℃） 1.9 [0℃~55℃] 2.2	放电时欠压超过报警阈值，将影响电池的循环寿命，用户放电不允许超过此电压值 When the under voltage exceeds the alarm threshold during discharge, it will affect the cycle life of the battery. The user discharge must not exceed this voltage value.	
最小工作电压 Minimum working voltage V	[-30℃~0℃） 2.0 [0℃~55℃] 2.5	正常工作的最小临界值 Minimum limit of normal operation	
电池温度过高一般报警阈值℃ Battery temperature is too high, general alarm threshold °C	55	电池工作温度超过此温度，将会限制电池使用功率 Battery operating temperature above this temperature will limit cell power	
电池温度过高严重报警阈值℃ Battery temperature is too high, critical alarm threshold °C	60	电池温度超过此报警阈值，将会影响电池安全性能，用户使用时不应超过此温度 If the cell temperature exceeds this alarm threshold, it will affect the cell safety performance. The user should not exceed this temperature when using it.	

电池温度过低一般报警阈值℃ Low battery temperature general alarm threshold ℃	-20	电池工作温度低于此温度，将会限制电池使用功率 If the operating temperature of the battery is lower than this temperature, the power of the battery will be limited	
电池温度过低严重报警阈值℃ Battery temperature is too low, critical alarm threshold °C	-30	电池温度低于此报警阈值，将会影响电池安全性能，用户使用时不应低于此温度 If the cell temperature is lower than this alarm threshold, it will affect the safety performance of the cell. The user should not use it below this temperature.	

A.2.2 Pack 设计提供的电池信息 Cell Information for Pack Design

电池型号 Cell Model	符号 Symbol	数值 Numerical value	意见 Suggestions	参考 Reference
Pack 装配过程最大允许压力 Maximum allowable pressure during Pack assembly	F	600kgf		17%±3%SOC
Pack 装配后电芯最终压力 Optimal pressure after Pack assembly	F	150±50kgf		17%±3%SOC

A.2.3 电芯结构相关信息 Information related to battery cell structure

铝壳 Aluminum shell	外形尺寸 External dimensions	22.2±0.5×284.8±0.5×120±0.5
	壁厚 Wall thickness	0.4mm
盖板设计 Cover plate design	端子高度 Terminal height	3.2mm(不含包胶及顶盖贴片) (Do not including adhesive and top cover patches)
	端子尺寸 Terminal size	52*15mm
	端子扭矩 Terminal torque	8N*m
	端子拉拔力 Terminal pulling force	≥800N
	防爆阀开启压力 Explosion proof valve opening pressure	0.6±0.15MPa
	基板厚度 Substrate thickness	1.5mm
	极柱中心距 Center to center distance of pole	/
	盖板周边焊 Welding around the cover plate	侧焊 Side welding

A.3 在选定条件下的循环寿命 Cycle Life at Certain Condition

常温循环寿命 Cycle Life at Certain Condition

测试条件 Test Conditions

充电：阶梯恒流充电 1C 充电至 3.5V，0.5C 充电至 3.60V，0.25C 充电至 3.62V，0.1C 充电至 3.65V

Charging: step constant current charging 1C to 3.5V，0.5C to 3.60V，0.25C to 3.62V，0.1C to 3.65V

放电：1C 放电至截止电压 2.5V；

Discharge: 1C discharge to 2.5V

温度：25℃；

Temperature: 25℃

静置时间：充电/放电后静置 60min。

Rest time: 60 min after charging/discharging

A.4　SOC-OCV

A4.1 充电 SOC-OCV charge SOC-OCV

T	-30℃	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
SOC	电压/V	电压/V	电压/V	电压/V	电压/V	电压/V	电压/V	电压/V
100%	3.3318	3.3325	3.3332	3.3339	3.3345	3.3358	3.3370	3.3380
99%	3.3325	3.3330	3.3335	3.3340	3.3344	3.3356	3.3361	3.3365
98%	3.3322	3.3326	3.3331	3.3335	3.3338	3.3349	3.3352	3.3353
97%	3.3316	3.3320	3.3324	3.3328	3.3331	3.3341	3.3342	3.3341
96%	3.3310	3.3314	3.3318	3.3322	3.3324	3.3334	3.3334	3.3332
95%	3.3306	3.3310	3.3313	3.3317	3.3319	3.3329	3.3328	3.3326
90%	3.3320	3.3322	3.3325	3.3327	3.3328	3.3335	3.3333	3.3329
85%	3.3314	3.3315	3.3317	3.3319	3.3320	3.3326	3.3324	3.3319
80%	3.3309	3.3310	3.3311	3.3313	3.3314	3.3318	3.3316	3.3310
75%	3.3312	3.3311	3.3312	3.3313	3.3314	3.3317	3.3316	3.3304
70%	3.3300	3.3301	3.3301	3.3303	3.3302	3.3306	3.3303	3.3297
65%	3.3296	3.3295	3.3296	3.3296	3.3297	3.3296	3.3297	3.3289
60%	3.2950	3.2962	3.2976	3.2991	3.3005	3.3017	3.3054	3.3062
55%	3.2863	3.2874	3.2888	3.2903	3.2916	3.2936	3.2970	3.2979
50%	3.2843	3.2853	3.2867	3.2883	3.2897	3.2915	3.2953	3.2962
45%	3.2841	3.2849	3.2861	3.2877	3.2891	3.2906	3.2946	3.2954
40%	3.2847	3.2853	3.2863	3.2877	3.2890	3.2901	3.2942	3.2950
35%	3.2855	3.2860	3.2868	3.2881	3.2894	3.2896	3.2938	3.2941
30%	3.2837	3.2831	3.2825	3.2820	3.2816	3.2833	3.2806	3.2793
25%	3.2696	3.2681	3.2668	3.2656	3.2648	3.2665	3.2627	3.2615
20%	3.2531	3.2518	3.2506	3.2495	3.2487	3.2493	3.2463	3.2448
15%	3.2277	3.2262	3.2248	3.2236	3.2226	3.2214	3.2190	3.2171
10%	3.2127	3.2114	3.2103	3.2093	3.2086	3.2052	3.2048	3.2029
5%	3.1686	3.1645	3.1604	3.1565	3.1530	3.1480	3.1395	3.1340
4%	3.1268	3.1224	3.1182	3.1140	3.1105	3.1048	3.0967	3.0924
3%	3.0726	3.0680	3.0636	3.0592	3.0555	3.0495	3.0409	3.0359
2%	3.0006	2.9958	2.9910	2.9864	2.9824	2.9763	2.9669	2.9615
1%	2.8988	2.8936	2.8885	2.8834	2.8790	2.8730	2.8623	2.8559
0%	2.7380	2.7336	2.7289	2.7240	2.7195	2.7161	2.7032	2.6940

A4.2 放电 SOC-OCV discharge SOC-OCV

T	-30℃	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
SOC	电压/V	电压/V	电压/V	电压/V	电压/V	电压/V	电压/V	电压/V
100%	3.3370	3.3375	3.3379	3.3384	3.3387	3.3399	3.3404	3.3412
99%	3.3376	3.3380	3.3385	3.3389	3.3392	3.3403	3.3406	3.3415
98%	3.3380	3.3383	3.3387	3.3392	3.3394	3.3404	3.3407	3.3416
97%	3.3372	3.3376	3.3380	3.3384	3.3387	3.3395	3.3397	3.3406
96%	3.3361	3.3365	3.3369	3.3373	3.3375	3.3384	3.3385	3.3392
95%	3.3364	3.3367	3.3371	3.3374	3.3376	3.3384	3.3383	3.3386
90%	3.3372	3.3376	3.3378	3.3381	3.3382	3.3388	3.3386	3.3392
85%	3.3378	3.3380	3.3383	3.3385	3.3386	3.3390	3.3388	3.3394
80%	3.3382	3.3384	3.3386	3.3388	3.3389	3.3391	3.3390	3.3396
75%	3.3386	3.3388	3.3389	3.3390	3.3391	3.3392	3.3391	3.3397
70%	3.3388	3.3389	3.3390	3.3392	3.3392	3.3391	3.3392	3.3398
65%	3.3376	3.3378	3.3381	3.3384	3.3387	3.3380	3.3388	3.3397
60%	3.3066	3.3077	3.3089	3.3102	3.3114	3.3124	3.3155	3.3177
55%	3.3009	3.3020	3.3033	3.3046	3.3058	3.3064	3.3101	3.3121
50%	3.2991	3.3002	3.3015	3.3029	3.3041	3.3041	3.3082	3.3101
45%	3.2979	3.2990	3.3002	3.3016	3.3028	3.3025	3.3069	3.3087
40%	3.2970	3.2980	3.2992	3.3006	3.3017	3.3012	3.3058	3.3076
35%	3.2963	3.2972	3.2983	3.2996	3.3007	3.3001	3.3045	3.3062
30%	3.2958	3.2964	3.2972	3.2982	3.2992	3.2983	3.3019	3.3032
25%	3.2939	3.2934	3.2929	3.2925	3.2922	3.2929	3.2909	3.2917
20%	3.2754	3.2744	3.2736	3.2728	3.2723	3.2719	3.2698	3.2701
15%	3.2459	3.2448	3.2438	3.2429	3.2422	3.2407	3.2391	3.2390
10%	3.2241	3.2231	3.2223	3.2216	3.2211	3.2171	3.2174	3.2169
5%	3.1959	3.1923	3.1888	3.1853	3.1825	3.1769	3.1701	3.1676
4%	3.1599	3.1561	3.1524	3.1487	3.1457	3.1399	3.1328	3.1301
3%	3.1141	3.1101	3.1062	3.1024	3.0993	3.0935	3.0860	3.0831
2%	3.0550	3.0508	3.0467	3.0427	3.0393	3.0334	3.0253	3.0221
1%	2.9732	2.9687	2.9643	2.9599	2.9562	2.9504	2.9411	2.9373
0%	2.8382	2.8332	2.8282	2.8231	2.8184	2.8148	2.8011	2.7940

A.5 不同温度和 SOC 条件下的放电 DCR Regeneration DCR at Different Temperature and SOC

DCR/mΩ @3C 10s							
T/SOC	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
10%	/	4.106	2.756	2.731	1.252	1.038	0.923
20%	/	3.569	2.558	2.153	1.124	0.909	0.822
30%	4.769	3.424	2.286	1.902	1.050	0.861	0.774
40%	4.603	3.150	2.098	1.769	1.001	0.826	0.758
50%	4.320	2.915	1.980	1.685	0.962	0.785	0.721
60%	4.043	2.759	1.904	1.672	0.963	0.748	0.675
70%	3.838	2.660	1.872	1.639	0.947	0.783	0.716
80%	3.702	2.600	1.826	1.591	0.919	0.758	0.693
90%	3.636	2.554	1.784	1.541	0.886	0.727	0.663
注：/表示不支持 10 s　　Note: / indicates that 10s							

A.6 不同温度和 SOC 条件下的回馈 DCR Regeneration DCR at Different Temperature and SOC

DCR/m Ω 10s							
T/SOC	-20℃ (0.33C)	-10℃ (0.33C)	0℃ (0.33C)	10℃ (2C)	25℃ (2.25C)	45℃ (2.25C)	55℃ (2.25C)
10%	12.968	8.457	5.964	1.784	1.080	0.965	0.843
20%	12.337	7.704	4.539	1.745	1.050	0.933	0.810
30%	10.758	6.685	3.270	1.725	1.020	0.905	0.783
40%	10.477	5.294	2.964	1.693	0.992	0.882	0.762
50%	8.831	4.870	2.761	1.655	0.960	0.856	0.739
60%	8.262	4.612	2.606	1.640	0.937	0.831	0.709
70%	7.960	4.432	2.508	1.623	0.930	0.856	0.746
80%	7.772	4.331	2.443	1.587	0.912	0.841	0.727
90%	7.787	4.313	2.388	1.543	0.880	0.806	0.689

A.7 不同温度和 SOC 脉冲放电峰值功率 Peak power pulse discharge at different temperatures and SOC

10s 脉冲放电峰值功率/W Peak power pulse discharge @10s /W								
T/SOC	-30℃	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
10%	151	358	591	811	1100	1898	1650	1056
20%	165	468	756	963	1513	2420	2173	1390
30%	193	536	921	1320	1650	2530	2296	1470
40%	206	619	1045	1554	1800	2600	2544	1628
50%	206	633	1141	1705	2000	3094	2805	1795
60%	220	646	1196	1843	2186	3094	2860	1830
70%	220	756	1348	1843	2228	3094	2929	1874
80%	248	784	1403	1843	2255	3163	2984	1910
90%	248	811	1444	1939	2269	3163	2984	1910
100%	275	894	1581	1980	2379	3300	3053	1954

30s 脉冲放电峰值功率/W Peak power pulse discharge @30s /W								
T/SOC	-30℃	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
10%	110	234	385	550	784	1334	1155	739
20%	138	316	495	646	1059	1691	1513	968
30%	138	358	605	908	1155	1774	1609	1030
40%	138	399	701	1059	1458	2049	1774	1135
50%	138	413	756	1169	1499	2173	1953	1250
60%	151	426	798	1251	1526	2173	1994	1276
70%	151	495	894	1251	1554	2173	2049	1311
80%	179	509	935	1320	1568	2214	2104	1346
90%	179	509	935	1320	1568	2214	2104	1346
100%	193	578	1045	1430	1664	2310	2145	1373

60s 脉冲放电峰值功率/W Peak power pulse discharge @60s /W								
T/SOC	-30℃	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
10%	69	124	193	261	426	715	701	449
20%	83	165	261	413	646	1100	1100	704
30%	110	289	481	729	935	1403	1416	906
40%	110	330	550	853	1114	1513	1444	924
50%	110	330	591	935	1196	1609	1471	942
60%	124	344	619	1004	1224	1691	1499	959
70%	124	399	715	1004	1251	1719	1513	968
80%	138	413	743	1059	1265	1760	1526	977
90%	138	413	743	1059	1265	1788	1540	986
100%	151	426	770	1155	1265	1815	1540	986

A.8　不同温度和 SOC 脉冲回馈峰值功率 Peak power pulse feedback at different temperatures and SOC

10s 脉冲回馈峰值功率/W Peak power pulse feedback@10s /W							
T/SOC	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
0%	250	510	852	1698	2700	2430	1652
10%	160	340	725	1507	2350	2115	1438
20%	138	300	656	1211	2000	1800	1224
30%	122	270	608	1024	1800	1620	1102
40%	120	260	550	948	1700	1530	1040
50%	118	250	492	861	1600	1440	979
60%	110	240	490	813	1500	1350	918
70%	93	200	427	742	1400	1260	857
80%	91	190	415	695	1350	1215	826
90%	89	180	402	630	1200	1080	734

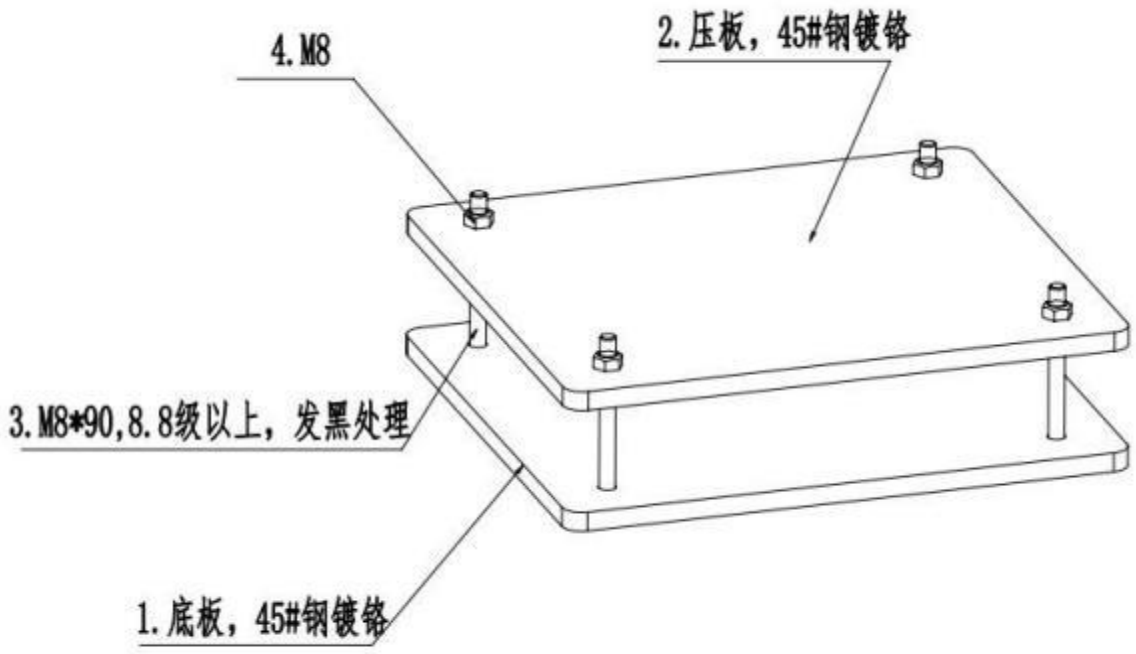
30s 脉冲回馈峰值功率/W Peak power pulse feedback@30s /W							
T/SOC	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
0%	223	620	743	1539	2100	1890	1285
10%	146	380	636	1436	1900	1710	1163
20%	127	300	591	1135	1650	1485	1010
30%	114	260	564	942	1600	1440	979
40%	113	250	492	868	1450	1305	887
50%	112	240	420	784	1480	1332	906
60%	98	230	419	725	1320	1188	808
70%	77	190	365	646	1280	1152	783
80%	74	180	360	592	1250	1125	765
90%	70	170	356	524	1200	1080	734

60s 脉冲回馈峰值功率/W Peak power pulse feedback@60s /W							
T/SOC	-20℃	-10℃	0℃	10℃	25℃	45℃	55℃
0%	213	510	653	1396	1900	1710	1163
10%	137	350	591	1247	1700	1530	1040
20%	121	280	551	1005	1600	1440	979
30%	111	250	528	853	1500	1350	918
40%	105	240	457	762	1400	1260	857
50%	100	220	387	664	1430	1287	875
60%	91	210	383	584	1300	1170	796
70%	74	180	331	488	1200	1080	734
80%	71	170	322	439	1120	1008	685
90%	69	160	313	380	1080	972	661

A.9 企业钢板夹具的使用 Using of steel plate

钢制夹板（45#钢镀铬，厚度 1cm）：夹具需覆盖住电芯大面，并做好绝缘处理，以免接触正负极导致电芯短路，夹板之间用4 个 M8 螺栓固定，夹具工装如下图所示：

Steel plate(45#, thickness of 1 cm):The plates should cover the cell, and pay attention to the insulation to prevent short circuits, the plates should to be secured with four M8 bolts, as shown below:



电芯夹具示意图 **Steel plateS**

IFP22290120A-96Ah 锂离子电池产品环保声明

Environmental declaration of IFP22290120A-96Ah Lithium Battery Product

根据欧盟(EU)2015/863 指令；2006-66-EC 电池指令要求，如下表所示 镉、铅、汞、六价铬、多溴联苯(PBB)、多溴二苯醚(PBDE)及邻苯二甲酸酯(PAEs)共计 10 种物质纳入禁用物质清单。

In accordance with EU directive 2015/863, battery instruction requirements in 2006-66-EC， as shown in the following table, a total of 10 substances are included in the RSL, for example Cadmium、Lead 、Mercury、Hexavalent chromium、Polybrominated biphenyls (PBB)、Polybrominated diphenyl ethers (PBDE) and phthalate (PAEs).

A. 12 十种禁用物质清单

A. 12 List of 10 restricted substances

RoHS 禁用物质 **RoHS restricted substance**	**最高限值（PPM）The highest limit（PPM）**	**说明** **Discription**
镉 Cadmium(Cd)	20	2006-66-EC 指令要求 2006-66-EC Directive Requirement
铅 Lead (Pb)	40	2006-66-EC 指令要求 2006-66-EC Directive Requirement
汞 Mercury (Hg)	5	2006-66-EC 指令要求 2006-66-EC Directive Requirement
六价铬 Hexavalent chromium (Cr6+)	1000	RoHS 1.0 已有限用物质 RoHS 1.0 limited substance
多溴联苯 Polybrominated biphenyls (PBB)	1000	RoHS 1.0 已有限用物质 RoHS 1.0 limited substance
多溴二苯醚 Polybrominated diphenyl ethers (PBDE)	1000	RoHS 1.0 已有限用物质 RoHS 1.0 limited substance
邻苯二甲酸二(2-乙基已基)酯 Diphthalate (2-ethylhexyl) ester （DEHP- Di(2-ethylhexyl)Phthalate）	1000	RoHS 2.0 新增限用物质 RoHS 2.0 limited substance
邻苯二甲酸丁苄酯 Benzyl butyl phthalate （BBP- Benzyl Butyl Phthalate）	1000	RoHS 2.0 新增限用物质 RoHS 2.0 limited substance
邻苯二甲酸二丁酯 Dibutyl phthalate （DBP-Di-n-butyl Phtalate）	1000	RoHS 2.0 新增限用物质 RoHS 2.0 limited substance
邻苯二甲酸二异丁酯 Diisobutyl phthalate （DIBP-Diiso butyl Phthalate）	1000	RoHS 2.0 新增限用物质 RoHS 2.0 limited substance

从发布日期立即实施执行 Execute immediately from release date.